Capstone 3 Project Report

Title

Forecasting Healthcare Call Demand Across the COVID-Era Structural Break: Evidence from Bangladesh Health Portal Call Data

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Executive Summary

This project developed time-series forecasting models to predict healthcare call demand using monthly Bangladesh Health Portal data spanning 2016–2025. Because the COVID-19 pandemic introduced a major structural break, multiple forecasting approaches were evaluated using expanding-window validation, including Seasonal Naïve, ETS (Holt–Winters), and ARIMA/SARIMA-family models. A log-transformed ARIMA(1,1,0) model achieved the strongest performance, reducing MAPE from approximately 75% for baseline seasonal approaches to 11.82% during validation and 9.92% on a held-out test set. Results suggest that recent temporal dynamics are more predictive than stable yearly seasonality in post-COVID healthcare demand forecasting.

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1. Problem

Healthcare systems rely on accurate demand forecasting to:

- allocate staff
- manage call center capacity
- maintain timely access to care

This project analyzes monthly call volume data from Bangladesh's national health portal to forecast future demand.

A key challenge is the COVID-19 structural break, which disrupted historical patterns in the series (Figure 1):

- sudden spikes in demand
- shifts in baseline call volume
- weakening of traditional seasonal patterns

Goal: Develop a forecasting model that remains accurate despite these disruptions.

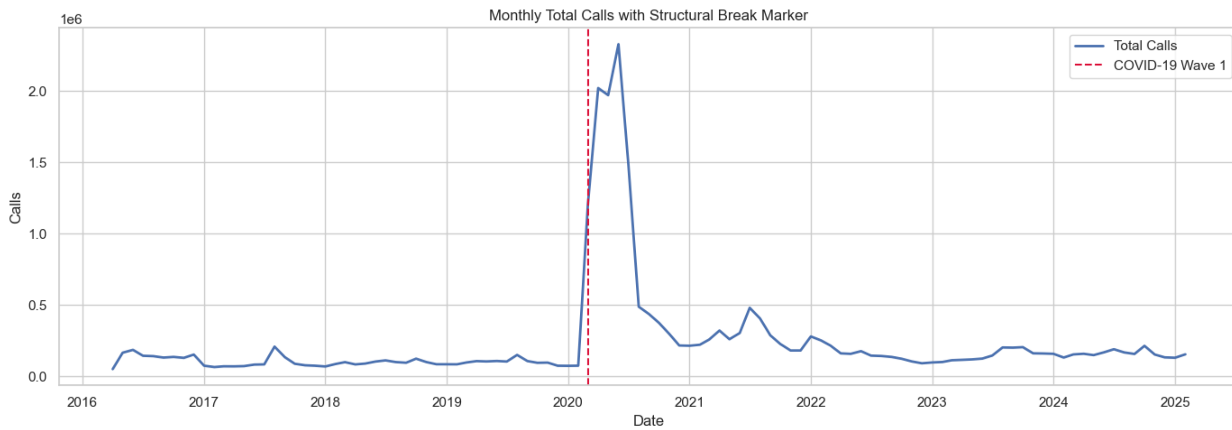

Figure 1. Monthly Total Calls with Structural Break Marker

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2. Data

The dataset used in this analysis captures monthly healthcare call volumes from Bangladesh's national health portal from 2016-04-01 to 2025-02-01.

- **Primary Source (Original Data)**
  Directorate General of Health Services (DGHS), Government of Bangladesh
  Healthcare Report
  http://16263.dghs.gov.bd/report/report.php
  (Accessed: March 18, 2025)

- **Accessed via Kaggle (Curated Dataset)**
  Basak, S. (2025). *Healthcare Call Data Analysis During Emergency Times*
  https://www.kaggle.com/datasets/shuvokumarbasak2030/healthcare-call-data-analysis-duringemergencytimes/data

The dataset was accessed via Kaggle but originates from the DGHS Bangladesh Health Portal. It reflects real-world healthcare system usage and captures demand fluctuations during public health emergencies, making it well-suited for evaluating forecasting models under structural disruption

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3. Approach

Preprocessing & Modeling Setup

The target variable (`total calls`) was log-transformed using `log1p` for ARIMA/SARIMA-family models to reduce right skew and stabilize variance following the large demand spikes observed during the COVID-19 period. Forecasts were converted back to the original call-count scale using expm1 prior to evaluation.

Seasonal Naïve and ETS models were fit on the raw call-count scale because these models rely directly on level and seasonal structure.

The 107-month time series was divided into three chronological subsets to preserve temporal structure and avoid data leakage:

1. **Initial training window:** 69 months (2016-04-01 to 2021-12-01)
2. **Evaluation window:** 24 months (2022-01-01 to 2023-12-01)
3. **Held-out test set:** 14 months (2024-01-01 to 2025-02-01)

The evaluation window was used for one-step-ahead expanding-window validation, allowing models to be assessed under realistic forecasting conditions in which only past information is available at each prediction step. A final held-out test set was reserved for unbiased evaluation of the selected model.

Feature Engineering

Additional time-series features were engineered during preprocessing, including:

- **Structural break indicator (`covid_period`)** to identify observations occurring during the COVID-era disruption
- **Cyclical month encoding (`month_sin, month_cos`)** to represent seasonal structure while preserving cyclical continuity between December and January

These features were created to support potential future extensions using exogenous or regression-based forecasting models, although the final ARIMA model relied solely on autoregressive structure and differencing.

Models Evaluated

Three classes of forecasting models were evaluated using one-step-ahead expanding-window validation (Table 1).

A **Seasonal Naïve (lag 12)** model was used as the baseline forecasting approach. This method predicts future values by repeating the observed value from the same month in the previous year. Although simple, Seasonal Naïve provides an important benchmark for determining whether more advanced models meaningfully improve upon basic seasonal repetition.

An **ETS (Holt–Winters exponential smoothing)** model was evaluated as a structured time-series forecasting approach incorporating level, trend, and seasonal components. An additive ETS specification with a damped trend and yearly seasonality (seasonal_periods=12) was selected to model gradual changes in healthcare call demand over time.

Finally, multiple **ARIMA/SARIMA-family models** were evaluated to capture autoregressive structure, differencing behavior, and potential seasonal dynamics. A focused grid search was conducted across six ARIMA and SARIMA parameter combinations using expanding-window cross-validation. Models were trained on a log-transformed version of the target variable to stabilize variance and reduce the influence of large COVID-era demand spikes.

The non-seasonal ARIMA models substantially outperformed both the Seasonal Naïve and ETS approaches, while SARIMA models failed to improve forecasting performance despite incorporating seasonal structure.

Model	**Purpose**
Seasonal Naïve	Baseline seasonal repetition model
ETS (Holt–Winters)	Level, trend, and seasonal smoothing
ARIMA / SARIMA	Autoregressive forecasting with differencing and optional seasonal structure

Table 1. Project Model Summary

Evaluation Strategy

Expanding-window validation was selected instead of random train-test splitting because random sampling violates temporal structure and can introduce data leakage in time-series forecasting problems. Models were evaluated using one-step-ahead expanding-window validation to simulate real-world forecasting conditions. At each evaluation step, models were trained using only historical observations available prior to the forecast date, preventing data leakage and preserving temporal structure.

After model selection, the best-performing model was refit using all pre-test observations and evaluated on a final held-out test set spanning January 2024 through February 2025.

Forecast performance was assessed using multiple complementary metrics:

- **MAE (Mean Absolute Error)** to measure average absolute forecast error
- **RMSE (Root Mean Squared Error)** to emphasize larger forecast errors
- **MAPE (Mean Absolute Percentage Error)** to evaluate relative percentage error
- **SMAPE (Symmetric Mean Absolute Percentage Error)** to provide a scale-independent error metric less sensitive to extreme values

4. Findings

Model Comparison

Table 2 summarizes validation performance across the primary forecasting approaches. Models were compared using multiple metrics, with model selection based primarily on MAPE due to the large changes in scale observed across the series. RMSE was also considered to evaluate the magnitude of large forecast errors.

Model	RMSE	MAPE %
ARIMA(1,1,0)	31,044.9	11.82
Seasonal Naïve (12)	126,489.2	74.42
ETS (Holt-Winters) additive	150,841	75.52

Table 2. Model Metrics

Multiple ARIMA-family models demonstrated very similar performance across metrics. ARIMA(1,1,0) was ultimately selected because it achieved the lowest MAPE while maintaining comparable RMSE with a simpler model structure. ARIMA dramatically outperformed Seasonal Naïve and ETS models (Figure 2).

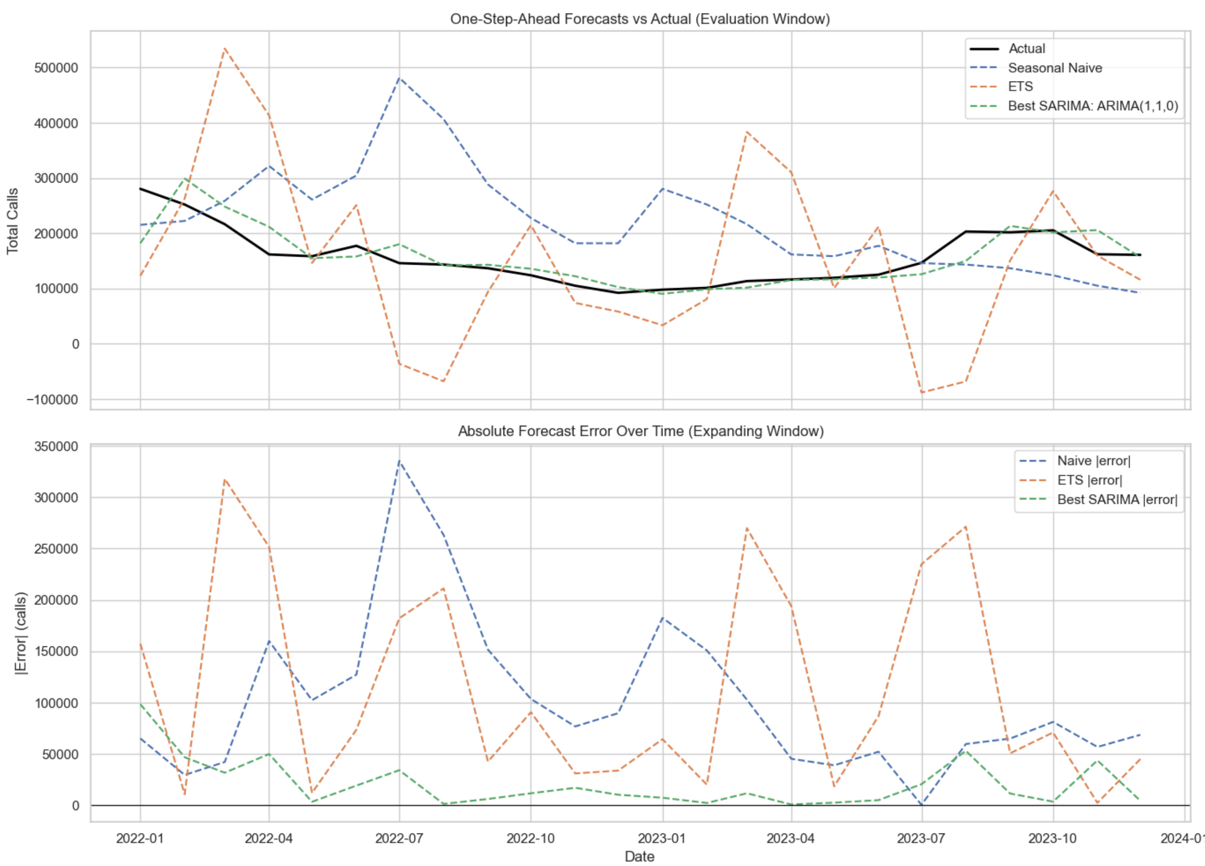

Figure 2. Top panel: One-step-ahead expanding-window forecasts compared with observed call volume. Bottom panel: Absolute forecast error over time.

On the final held-out test set the ARIMA(1,1,0) model performed even better with RMSE = 21,736.9 and MAPE % = 9.92. This consistency suggests the model generalizes well and effectively captures the underlying dynamics of the series. A zoomed-in view of the final held-out test forecast preceded by a year of training data for context is shown in Figure 3.

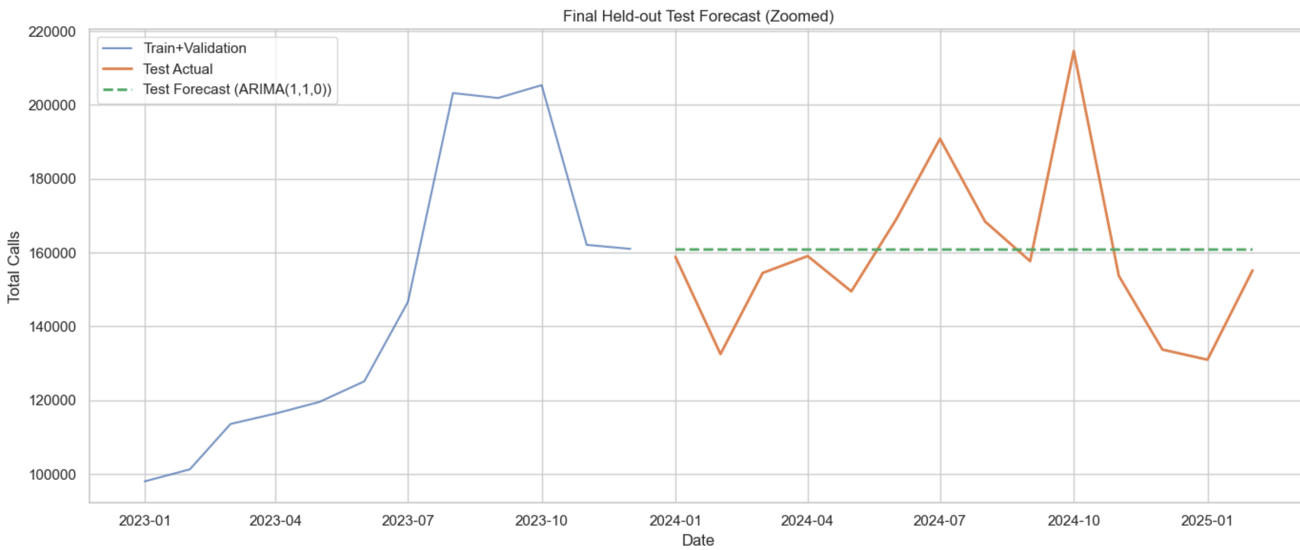

Figure 3. Final Held-Out Test Forecast (Zoomed).

Key Insights

1. Seasonality is weak or unstable: Seasonal Naïve and ETS performed poorly, indicating that yearly patterns are not reliable post-COVID.
2. Short-term dynamics dominate: ARIMA captures recent behavior effectively, suggesting that demand is driven by recent trends, not repeating cycles.
3. Structural break matters: Since COVID fundamentally changed the series, models assuming stable historical patterns underperform.
4. Forecast behavior: ARIMA produces relatively flat forecasts, which indicates that future demand is expected to remain near current levels unless new shocks occur.

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5. Recommendations

1. The ARIMA-based forecast, which achieved ~10% error, is suitable for monthly staffing, call center capacity planning, and short-term operational forecasting.

2. Avoid relying on historical seasonality alone, since seasonal patterns are unstable.

3. Monitor for structural changes and retrain regularly (e.g. quarterly) as demand is highly sensitive to external events.

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6. Future Work

1. Explore hybrid or ML models to combine time features and external drivers, such as public health events.
2. Explicitly address the COVID-19 structural break with regime-switching models and change-point detection.
3. Evaluate multi-step forecasting strategies to assess longer forecasting horizons beyond one-step-ahead prediction.

7. Conclusion

Overall, the results demonstrate that relatively simple autoregressive models can provide highly effective healthcare demand forecasts when paired with appropriate preprocessing and evaluation strategies. The final ARIMA model substantially outperformed traditional seasonal approaches and provides a practical framework for short-term healthcare operations planning under structurally changing conditions.